Software Requirements Specification (SRS)

Passport Appointment System

Subject: Software Engineering · **Assignment:** 1

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This document follows the structure recommended by IEEE Std 830-1998, *Recommended Practice for Software Requirements Specifications*.

Document detail	Value
Document title	SRS for Passport Appointment System
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Prepared by	Team of four students (listed above)

1. Introduction

1.1 Purpose

This SRS describes the functional and non-functional requirements of the Passport Appointment System (PAS), version 1.0. It is written for the development team, testers, project guide and the evaluators, and serves as the agreed basis for design, implementation and testing.

1.2 Scope

PAS is a web-based application that lets citizens apply for a passport, pay the fee online and book, reschedule or cancel an appointment at a Passport Seva Kendra (PSK) or Post Office Passport Seva Kendra (POPSK). Officers use it to view daily schedules and record

document verification. Administrators use it to manage centres, slots and holidays.

The system will:

- register and authenticate users;
- capture passport application details and uploaded documents;
- calculate and collect the application fee through a payment gateway;
- show available slots and prevent double booking;
- send confirmation and reminder notifications;
- give officers and administrators schedules and reports.

The system will not handle police verification, passport printing and dispatch, or biometric device integration.

1.3 Definitions, Acronyms and Abbreviations

Term	Meaning
PAS	Passport Appointment System
PSK / POPSK	Passport Seva Kendra / Post Office Passport Seva Kendra
SRS	Software Requirements Specification
Applicant	A citizen who applies for a passport
Slot	A fixed time window at a centre with a limited capacity
ARN	Application Reference Number issued after submission
OTP	One-Time Password
Tatkal	Fast-track passport service with a higher fee
HTTPS	Hypertext Transfer Protocol Secure
RBAC	Role-Based Access Control

1.4 References

- IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.
- Pressman, R. S., *Software Engineering: A Practitioner's Approach*.

- Sommerville, I., *Software Engineering*.
- Passport Seva portal, Ministry of External Affairs, Government of India (general process reference).

1.5 Overview

Section 2 describes the product in general terms. Section 3 lists the specific, testable requirements. Section 4 holds appendices and a traceability summary.

2. Overall Description

2.1 Product Perspective

PAS is a new, self-contained web application with a three-tier structure (user interface, business logic, database). It connects to two external systems: a payment gateway for fee collection and an SMS and email service for notifications.

External system	Purpose	Interaction
Payment Gateway	Process online fee payment	Request payment, receive success or failure status
SMS / Email Service	Send notifications	Send message, receive delivery status

2.2 Product Functions

1. User registration, login and profile management.
2. Online passport application with draft saving and document upload.
3. Fee calculation and online payment.
4. Centre, date and slot search.
5. Appointment booking, rescheduling and cancellation.
6. Appointment receipt generation.
7. Notifications and reminders.
8. Officer schedule view and appointment status update.
9. Administration of centres, slots, holidays and officers.
10. Report generation.

2.3 User Characteristics

User class	Characteristics
Applicant	General public with basic internet skills, often on a mobile phone; varied education levels
Passport Officer	Trained government staff, uses the system daily on a desktop
Administrator	Technical staff with authority to configure the system

2.4 Constraints

- The system must run in a standard web browser (latest two versions of Chrome, Firefox, Edge and Safari).
- Personal data must be protected as required by Indian data protection law.
- Payments must go only through an approved payment gateway.
- The database must be relational.
- Fee amounts and booking limits must follow current government rules and be configurable.

2.5 Assumptions and Dependencies

- Applicants have a valid mobile number or email address and internet access.
- The payment gateway and SMS service are available during working hours.
- Centre working days, holidays and slot capacities are provided by the administrator.
- Scanned documents are supplied in PDF or JPEG format.

2.6 Apportioning of Requirements

Police verification, passport printing, dispatch tracking and a native mobile app are deferred to later versions.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

ID	Requirement
UI-1	The interface shall be responsive and usable on screens from 360 px wide phones to desktops.
UI-2	The booking flow shall be shown as steps (Register, Apply, Pay, Book, Receipt) with a progress indicator.
UI-3	Slot selection shall use a calendar that marks holidays and full dates.
UI-4	Errors shall be shown next to the field in plain language.
UI-5	The interface shall support English and Hindi.

3.1.2 Hardware Interfaces

PAS needs no special hardware. Users need a device with a web browser and internet connection. The server runs on a standard cloud or on-premise server.

3.1.3 Software Interfaces

Interface	Description
Payment Gateway API	HTTPS REST calls to initiate payment and receive a signed status response
SMS / Email API	HTTPS calls to send templated messages
Database	Relational database accessed by the business layer only

3.1.4 Communication Interfaces

All client-server and server-to-external traffic shall use HTTPS with TLS 1.2 or higher.

3.2 Functional Requirements

3.2.1 User Management

ID	Requirement	Priority
FR-1	The system shall let a new user register with name, date of birth, email, mobile number and password.	High
FR-2	The system shall verify the email or mobile number using an OTP before activating the account.	High
FR-3	The system shall authenticate users with login ID and password and lock the account for 15 minutes after 5 failed attempts.	High
FR-4	The system shall let a user reset a forgotten password through an OTP.	Medium
FR-5	The system shall allow access to features according to the user's role (Applicant, Officer, Administrator).	High

3.2.2 Application Handling

ID	Requirement	Priority
FR-6	The system shall let an applicant fill in the application form and save it as a draft.	High
FR-7	The system shall validate mandatory fields and formats before submission.	High
FR-8	The system shall let the applicant upload supporting documents (PDF or JPEG, maximum 2 MB each).	High
FR-9	The system shall issue a unique Application Reference Number (ARN) on submission.	High
FR-10	The system shall let the applicant track the application status using the ARN.	Medium

3.2.3 Fee and Payment

ID	Requirement	Priority
FR-11	The system shall calculate the fee from the application type (fresh, re-issue, Tatkal) and booklet size.	High
FR-12	The system shall redirect the applicant to the payment gateway and record the returned status.	High
FR-13	The system shall allow appointment booking only after a successful payment.	High
FR-14	The system shall issue a payment receipt and store the transaction reference.	Medium
FR-15	The system shall flag a payment that succeeded but has no booking, for refund or reconciliation.	Medium

3.2.4 Appointment Management

ID	Requirement	Priority
FR-16	The system shall display centres and available dates and slots.	High
FR-17	The system shall reserve a slot for an applicant and reduce its remaining capacity by one.	High
FR-18	The system shall prevent two applicants from taking the last place in a slot, including under simultaneous requests.	High
FR-19	The system shall allow only one active appointment per application.	High
FR-20	The system shall let an applicant reschedule an appointment up to 2 times, at least 24 hours before the slot.	Medium
FR-21	The system shall let an applicant cancel an appointment and shall release the slot.	Medium
FR-22	The system shall generate a printable receipt with the ARN, centre, date, time and a list of documents to bring.	High

3.2.5 Notifications

ID	Requirement	Priority
FR-23	The system shall send a confirmation by SMS and email after booking.	High
FR-24	The system shall send a reminder 24 hours before the appointment.	Medium
FR-25	The system shall notify the applicant of rescheduling, cancellation or centre closure.	Medium
FR-26	The system shall retry a failed notification up to 3 times.	Low

3.2.6 Officer Functions

ID	Requirement	Priority
FR-27	The system shall show an officer the day's appointments for the officer's centre.	High
FR-28	The system shall let an officer mark each appointment as Completed, Missed or Documents Incomplete.	High
FR-29	The system shall let an officer view the application and documents of an applicant with an appointment.	High

3.2.7 Administrator Functions

ID	Requirement	Priority
FR-30	The system shall let an administrator add, edit and deactivate centres.	High
FR-31	The system shall let an administrator define working days, holidays, slot length and capacity.	High
FR-32	The system shall let an administrator create and deactivate officer accounts.	High

ID	Requirement	Priority
FR-33	The system shall let an administrator generate reports of bookings, cancellations and no-shows by centre and date range.	Medium

3.3 Use Case Summary

Use case	Actor	Short description
UC-1 Register and Login	Applicant	Create an account and sign in
UC-2 Submit Application	Applicant	Fill the form, upload documents, receive ARN
UC-3 Pay Fee	Applicant, Payment Gateway	Pay the calculated fee online
UC-4 Book Appointment	Applicant	Choose centre, date and slot and receive a receipt
UC-5 Reschedule or Cancel	Applicant	Change or cancel an existing appointment
UC-6 Manage Daily Schedule	Officer	View appointments and update status
UC-7 Configure Centres and Slots	Administrator	Maintain centres, holidays and capacity
UC-8 Generate Reports	Administrator	Produce booking statistics

Use Case UC-4: Book Appointment

Field	Description
Actor	Applicant
Precondition	Applicant is logged in, the application is submitted and the fee is paid

Field	Description
Main flow	1. Applicant selects Book Appointment. 2. System lists centres. 3. Applicant picks centre and date. 4. System shows free slots. 5. Applicant selects a slot. 6. System reserves the slot, creates the appointment and shows the receipt. 7. System sends a confirmation.
Alternate flow	If the slot is taken meanwhile, the system shows the remaining slots and asks the applicant to choose again.
Exception flow	If payment is not verified, the system blocks booking and asks the applicant to complete payment.
Postcondition	Appointment is saved with status Scheduled and the slot capacity is reduced by one.

3.4 Performance Requirements

ID	Requirement
PR-1	Slot search results shall appear within 3 seconds for 95 percent of requests.
PR-2	The system shall support at least 1,000 concurrent users.
PR-3	A booking confirmation shall complete within 5 seconds after the applicant selects a slot.
PR-4	Pages other than payment shall load within 4 seconds on a 4G connection.
PR-5	The database shall hold at least 5 years of appointment records without slowing searches.

3.5 Logical Database Requirements

Entity	Main attributes
User	user_id, name, dob, email, mobile, password_hash, role

Entity	Main attributes
Application	application_id, user_id, type, status, arn, created_on
Payment	payment_id, application_id, amount, status, txn_ref
Centre	centre_id, name, city, address
Slot	slot_id, centre_id, date, start_time, end_time, capacity, booked_count
Appointment	appointment_id, application_id, slot_id, status, reference_no
Notification	notification_id, appointment_id, channel, message, sent_on

A slot belongs to one centre; an appointment belongs to one slot and one application; an application has at most one active appointment.

3.6 Design Constraints

- The system shall follow a layered architecture with separate presentation, business and data layers.
- Passwords shall be stored only as salted hashes.
- The system shall comply with applicable data protection and IT laws and with the standards of the issuing authority.

3.7 Software System Attributes

3.7.1 Reliability

- Slot booking shall run as a single transaction so a failure never leaves a half-booked slot.
- Data shall be backed up daily and the system shall recover from a failure within 4 hours.

3.7.2 Availability

- The system shall be available at least 99.5 percent of the time during working hours (8 am to 8 pm).

3.7.3 Security

- All sensitive data shall be encrypted in transit (TLS) and personal identifiers shall be encrypted at rest.
- Access shall be controlled by role (RBAC); an applicant can see only their own records.
- Login and booking pages shall use CAPTCHA to limit automated booking.
- Sessions shall expire after 15 minutes of inactivity.
- The system shall keep an audit log of logins, bookings, cancellations and administrative changes.

3.7.4 Maintainability

- The code shall be modular, with fee rules and booking limits held in configuration, not in code.
- Each module shall have unit tests covering its main paths.

3.7.5 Portability

- The application shall run on any major browser and on Windows or Linux servers without code changes.

3.7.6 Usability

- A first-time applicant shall be able to book an appointment within 10 minutes without help.
- The interface shall meet WCAG 2.1 level AA contrast and keyboard navigation guidelines.

3.8 Other Requirements

- **Legal:** the system shall show a privacy notice and obtain consent before storing personal data.
- **Localisation:** dates shall use the DD/MM/YYYY format and amounts shall be shown in rupees.

4. Appendices

4.1 Requirement Priority Scale

Priority	Meaning
High	Essential; the system cannot be released without it

Priority	Meaning
Medium	Important; should be included in version 1.0
Low	Desirable; may be deferred

4.2 Traceability Summary

Use case	Related requirements
UC-1 Register and Login	FR-1 to FR-5
UC-2 Submit Application	FR-6 to FR-10
UC-3 Pay Fee	FR-11 to FR-15
UC-4 Book Appointment	FR-16 to FR-19, FR-22, FR-23
UC-5 Reschedule or Cancel	FR-20, FR-21, FR-25
UC-6 Manage Daily Schedule	FR-27 to FR-29
UC-7 Configure Centres and Slots	FR-30 to FR-32
UC-8 Generate Reports	FR-33

4.3 Open Issues

- Final fee table and booking limits are to be confirmed against current government rules.
- The payment gateway vendor has not yet been chosen.
- Languages beyond English and Hindi are not yet decided.